

There is a push for higher ethanol blends. What is the impact on existing vehicles?

Higher ethanol blends such as E85 are not meant for use in existing conventional vehicles. They are intended for flex fuel vehicles (FFVs), which are specifically designed to handle higher ethanol content in fuel. While ethanol has lower calorific value than petrol, it also has higher octane value.

FFV engines are designed with suitable compression ratio and engine calibration to utilise this advantage, thereby reducing the impact on fuel economy. In future, hybrid FFV platforms may further help improve overall efficiency.

The government has already constituted a Working Group on FFVs with participation from oil marketing companies and vehicle manufacturers. Therefore, the introduction of FFVs and E85 fuel is expected to be calibrated and well-planned.



What is IOCL's roadmap for ethanol production from agricultural waste?

India's ethanol blending programme began with feedstocks such as sugarcane molasses and surplus grains because those supply chains already existed at scale.



“MARKET MOVING TO 2G ETHANOL”

A S Sahney, CMD of IOCL, says higher ethanol blends are for specific vehicles designed to handle the E85/E100 fuel. Edited excerpts from an interview with Aishwarya Baban Patil

However, the long-term direction is clearly moving towards second-generation (2G) ethanol made from agricultural residues such as paddy straw, corn stalks, bamboo, and other crop waste. The encouraging part is that India is no longer discussing this only at the policy level — commercial plants are already operational.

IndianOil has commissioned India's first commercial-scale 2G ethanol refinery at Panipat using paddy straw as feedstock. The project demonstrates that agricultural waste can be converted into

clean transport fuel while also addressing the major environmental problem of stubble burning in North India.



What is the strategy for the sustainable aviation fuel?

Indian Oil has already participated in India's first commercial passenger flight using indigenously produced SAF-blended ATF, supplied by Indian Oil and used by AirAsia in demonstration mode.

India's SAF roadmap is supported by abundant domestic feedstocks

such as used cooking oil, agri residue and biomass waste, which can gradually support aviation blending requirements.

For airline partners such as Akasa Air, AirIndia and others, the challenge is less about intent and more about scaling production economically. Global SAF production is still very limited, accounting for less than 1% of jet fuel demand.

IndianOil's approach is, therefore, to gradually expand domestic SAF capacity through refinery integration, biofuel partnerships, and indigenous technology development, ensuring that initial blending targets remain achievable while building long-term supply reliability for Indian carriers.



What is the SAF technology being explored by the company?

Indian Oil is pursuing a portfolio approach for SAF as the sector will require multiple pathways depending on feedstock availability, technology maturity, refinery integration and commercial viability.

IndianOil is also evaluating next-generation pathways such as alcohol-to-jet and biomass-based fuels. IndianOil will progressively integrate the most suitable SAF pathway at locations where feedstock access, hydrogen availability, refinery configuration and commercial fit are the strongest.